MODEL LAB · 2026-10-12

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

28,490
held-out plate appearances

3.05%
held-out HR rate

0.5961
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-01, scored on the 28,490 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5779	1.61×	0.02952	0.13559	0.00491
KNN comps Prior is the pooled rate of the nearest comparables.	0.5961	1.45×	0.02948	0.13468	0.00474
SVR Support-vector regression on the same feature matrix.	0.5887	1.43×	0.02953	0.13539	0.00646
Random forest Forest regression; a learned similarity metric.	0.5898	1.45×	0.02951	0.13508	0.00618
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5961	1.45×	0.02948	0.13468	0.00474
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5960	1.47×	0.02948	0.13468	0.00479
Linear regression Ordinary least squares on the same features.	0.5904	1.42×	0.02951	0.13512	0.00532
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	1.50×	0.02952	0.13522	0.00651
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5899	1.46×	0.02951	0.13512	0.00662
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5907	1.42×	0.02952	0.13520	0.00644

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.90** SVR **0.00** RF **0.10**

SHRINKAGE K PER PRIOR

league **249** knn **249** svr **249** rf **249** core **249** ensemble **249** linear **249** logistic **249** nn **249** xgb **249**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,116 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5823	1.48×	0.02319	0.11242	0.01187
KNN comps Prior is the pooled rate of the nearest comparables.	0.5746	1.08×	0.02323	0.11256	0.01296
SVR Support-vector regression on the same feature matrix.	0.5684	1.21×	0.02342	0.11561	0.01853
Random forest Forest regression; a learned similarity metric.	0.5740	1.35×	0.02334	0.11344	0.01619
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5746	1.08×	0.02323	0.11256	0.01296
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	0.94×	0.02324	0.11250	0.01259
Linear regression Ordinary least squares on the same features.	0.5766	0.94×	0.02333	0.11448	0.01600
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5846	1.08×	0.02332	0.11370	0.01486
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5721	1.21×	0.02329	0.11362	0.01393
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5843	1.21×	0.02337	0.11380	0.01667

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 17.5%, Junior Caminero 17.1%, Jackson Chourio 16.0%, Teoscar Hernández 15.9%, Freddie Freeman 14.7%, Jake Bauers 14.7%
Flat league	5/6	Junior Caminero 16.2%, Jackson Chourio 15.9%, Shohei Ohtani 15.9%, Teoscar Hernández 15.1%, Freddie Freeman 14.3%, Jo Adell 13.9%
KNN comps	6/6	Shohei Ohtani 17.5%, Junior Caminero 16.8%, Jackson Chourio 16.4%, Teoscar Hernández 16.0%, Jake Bauers 14.9%, Freddie Freeman 14.4%
SVR	5/6	Junior Caminero 18.5%, Shohei Ohtani 17.3%, Jackson Chourio 16.6%, Teoscar Hernández 15.9%, Jo Adell 14.9%, Jake Bauers 14.8%
Random forest	5/6	Shohei Ohtani 17.7%, Junior Caminero 17.5%, Jackson Chourio 16.7%, Teoscar Hernández 16.1%, Jake Bauers 14.9%, Jo Adell 14.5%
Core (KNN+SVR)	6/6	Shohei Ohtani 17.5%, Junior Caminero 16.8%, Jackson Chourio 16.4%, Teoscar Hernández 16.0%, Jake Bauers 14.9%, Freddie Freeman 14.4%
Forest blend	6/6	Shohei Ohtani 17.5%, Junior Caminero 16.9%, Jackson Chourio 16.4%, Teoscar Hernández 16.0%, Jake Bauers 14.9%, Freddie Freeman 14.4%
Linear regression	6/6	Shohei Ohtani 17.4%, Junior Caminero 17.0%, Jackson Chourio 16.4%, Teoscar Hernández 16.1%, Jake Bauers 14.7%, Freddie Freeman 14.6%
Logistic regression	5/6	Shohei Ohtani 17.8%, Junior Caminero 17.4%, Jackson Chourio 16.5%, Teoscar Hernández 16.1%, Jo Adell 14.9%, Jake Bauers 14.6%
Neural net (torch)	5/6	Shohei Ohtani 17.8%, Junior Caminero 17.5%, Jackson Chourio 16.6%, Teoscar Hernández 16.1%, Jo Adell 14.9%, Jake Bauers 14.8%
XGBoost	5/6	Shohei Ohtani 18.0%, Junior Caminero 17.9%, Jackson Chourio 16.8%, Teoscar Hernández 16.0%, Jake Bauers 15.0%, Jo Adell 14.7%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Shohei Ohtani**	LAD	**17.5%**	15.9%	17.5%	17.3%	17.7%	17.5%	17.5%	17.4%	17.8%	
2	**Junior Caminero**	TB	**17.1%**	16.2%	16.8%	18.5%	17.5%	16.8%	16.9%	17.0%	17.4%	
3	**Jackson Chourio**	MIL	**16.0%**	15.9%	16.4%	16.6%	16.7%	16.4%	16.4%	16.4%	16.5%	
4	**Teoscar Hernández**	LAD	**15.9%**	15.1%	16.0%	15.9%	16.1%	16.0%	16.0%	16.1%	16.1%	
5	**Freddie Freeman**	LAD	**14.7%**	14.3%	14.4%	14.3%	14.3%	14.4%	14.4%	14.6%	14.4%	
6	**Jake Bauers**	MIL	**14.7%**	13.7%	14.9%	14.8%	14.9%	14.9%	14.9%	14.7%	14.6%	
7	**Jo Adell**	CLE	**14.6%**	13.9%	14.4%	14.9%	14.5%	14.4%	14.4%	14.5%	14.9%	
8	**Ryan Vilade**	TB	**14.4%**	13.7%	14.3%	14.0%	14.5%	14.3%	14.3%	14.0%	13.9%	
9	**William Contreras**	MIL	**13.7%**	13.6%	13.8%	13.6%	13.7%	13.8%	13.8%	13.6%	13.6%	
10	**Jonathan Aranda**	TB	**13.6%**	12.8%	13.8%	13.2%	13.4%	13.8%	13.7%	13.6%	13.4%	
11	**Brice Turang**	MIL	**13.5%**	13.8%	13.2%	13.6%	13.2%	13.2%	13.2%	12.9%	12.8%	
12	**Mookie Betts**	LAD	**13.5%**	13.5%	13.4%	13.9%	13.4%	13.4%	13.4%	13.6%	13.6%	
13	**Will Smith**	LAD	**13.5%**	13.2%	13.6%	13.3%	13.5%	13.6%	13.6%	14.0%	13.8%	
14	**Yandy Díaz**	TB	**12.9%**	13.3%	12.9%	13.4%	13.3%	12.9%	12.9%	13.2%	13.0%	
15	**Christian Yelich**	MIL	**12.4%**	12.6%	12.9%	12.0%	12.6%	12.9%	12.9%	12.9%	12.3%	
16	**Victor Mesa Jr.**	TB	**12.3%**	11.9%	12.3%	12.9%	12.4%	12.3%	12.4%	12.5%	12.4%	
17	**Garrett Mitchell**	MIL	**11.8%**	11.2%	11.6%	10.5%	10.9%	11.6%	11.5%	11.0%	10.9%	
18	**Andy Pages**	LAD	**11.2%**	10.7%	10.9%	10.7%	10.7%	10.9%	10.9%	10.7%	10.6%	
19	**Kyle Tucker**	LAD	**10.8%**	10.9%	10.9%	10.7%	10.8%	10.9%	10.9%	10.7%	10.5%	
20	**Patrick Bailey**	CLE	**10.8%**	10.6%	11.2%	10.9%	10.7%	11.2%	11.1%	10.9%	10.5%	
21	**Nathaniel Lowe**	CLE	**10.4%**	10.1%	10.3%	10.2%	10.4%	10.3%	10.3%	10.2%	10.2%	

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
22	**Chase DeLauter** CLE	**10.2%**	10.3%	10.2%	10.0%	10.1%	10.2%	10.2%	10.0%	9.9%	
23	**Travis Bazzana** CLE	**10.0%**	10.0%	9.9%	9.6%	10.0%	9.9%	9.9%	9.8%	9.6%	
24	**José Ramírez** CLE	**9.6%**	9.6%	9.4%	9.6%	9.4%	9.4%	9.4%	9.7%	9.9%	
25	**Jonny DeLuca** TB	**8.8%**	9.6%	8.7%	8.8%	8.0%	8.7%	8.6%	8.6%	8.9%	
26	**Nick Fortes** TB	**8.4%**	9.3%	8.2%	7.9%	7.8%	8.2%	8.2%	8.7%	8.6%	
27	**Tommy Edman** LAD	**7.7%**	8.3%	7.5%	7.6%	7.2%	7.5%	7.5%	7.7%	7.6%	
28	**Petey Halpin** CLE	**7.3%**	8.4%	7.6%	8.0%	7.8%	7.6%	7.6%	8.5%	8.0%	
29	**Sal Frelick** MIL	**6.5%**	8.1%	6.6%	6.5%	6.1%	6.6%	6.5%	6.4%	6.4%	
30	**Brayan Rocchio** CLE	**6.3%**	6.8%	6.1%	6.3%	5.9%	6.1%	6.1%	6.0%	6.0%	
31	**David Hamilton** MIL	**6.1%**	7.7%	6.3%	5.8%	5.7%	6.3%	6.2%	6.0%	5.9%	
32	**Cooper Pratt** MIL	**6.1%**	7.9%	6.0%	5.3%	5.8%	6.0%	6.0%	5.4%	5.5%	
33	**Taylor Walls** TB	**6.0%**	8.1%	6.2%	5.3%	5.2%	6.2%	6.1%	6.3%	6.3%	
34	**Ben Williamson** TB	**5.9%**	9.3%	6.3%	5.8%	6.4%	6.3%	6.4%	4.2%	6.1%	
35	**Steven Kwan** CLE	**4.6%**	7.6%	5.3%	3.2%	3.9%	5.3%	5.2%	3.2%	4.6%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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